Data Analytics Assignment 6

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1 Exploratory Data Analysis

The first step of EDA was to look at what data was available. The dataset I used the student performance dataset found here: https://archive.ics.uci.edu/dataset/856/higher+education+students+performance+evaluation. The initial scan over the data revealed no missing data in any column, making cleaning much simpler. The columns did not however have proper labeling and thus had to be labeled before further exploration could occur. While there was no explicit missing data, some of the questions asked had a response that could be classified as similar to a missing value. This included the preparation for midterms column having a not applicable value of 3. Going forward if I plan to use this value for some part of the prediction it is important to take note and filter out these values. After running a shapiro test to find normal distributions in the dataset none were found. This is expected however as the dataset is relatively small and not exactly naturally occurring. Plotting some of the variables can give insight into future development models.

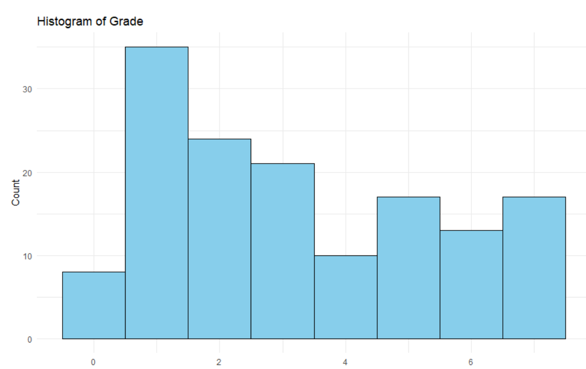

Figure 1: Grades Histogram

Grades is a good variable to create models for. There are a range of values to predict and fairly varied within the values themselves. General clustering could also be useful to see what hidden groups exist within the data and if they can be tied back to academic performance. As for the models used for prediction, any model seems to look like they will work well. The Grade values are already binned and are numeric so classification and regression are both strong options.

2 Model Development

2.1 Kmeans clustering

The first model developed was a clustering model. This model did not perform well or provide any insight to the data. Below is a photo of the clustering with each dot being a student and their corresponding grade.

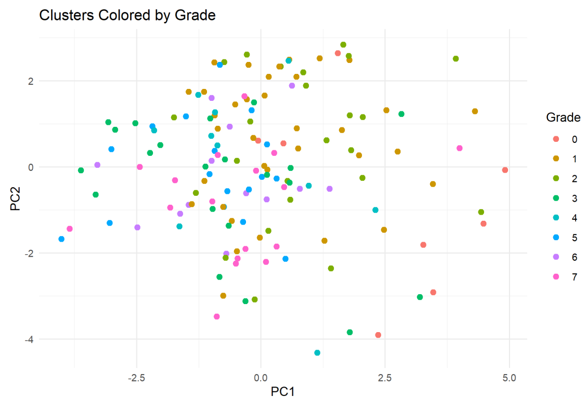

Figure 2: Clustering with grade colors

After this kmeans was used after removing the student id and course id columns and can be pictured below.

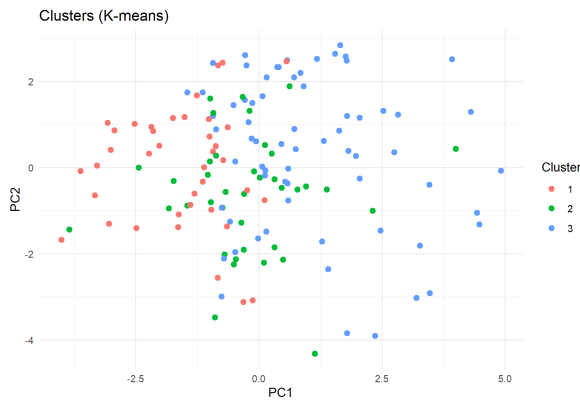

Figure 3: Kmeans Clustering

The figure shows, even at 3 clusters there is quite a bit of overlap and no real hidden subgroups can be found.

2.2 Linear Model

The second model developed was a linear model to predict grades. For this model I used the following variables: Listening during class, Weekly study hours, taking notes, and academic reading frequency.

I chose this model because I was curious if any of these factors had a major impact on overall grade results. The model itself did poorly, with the residual standard error being 2.217. This may have been small if grades were between 0 and 100, but in this dataset they are binned into categories, 1-8. On top of this none of the four predictors were significant. When looking at the figure below, it can be seen the predicted grade is all over the place and the findings from above are seen.

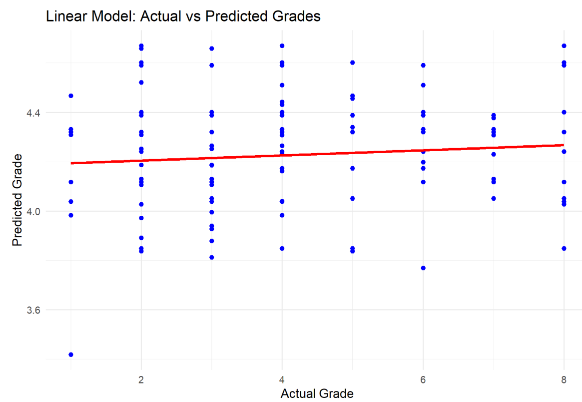

Figure 4: Linear Model

2.3 Random Forest

The third model developed was a random forest model again to predict grades. To compare it with the linear model I kep the same input independent variables. Again the model performed poorly, with a residual standard error of 2.053882. While this is slightly better than the linear model, I would not consider it a good performance by any means. As seen in the figure below the model does very poorly, with the predicted values being all over the place. Going forward the inputs to the model should change because these seem to have no significance to predicting the end grades.

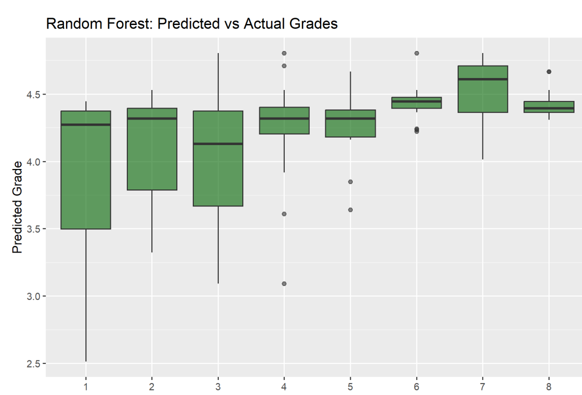

Figure 5: Random Forest Model

2.4 SVM with PCA

The last model developed is a SVM model utilizing dimensionality reduction and PCAs. The reason SVM was chosen was because it is another model that we covered in class and PCAs were used to help boost the models performance. As seen in the previous models, they performed poorly with the given independent variables. Because of this, the dimensionality reduction helps simplify the roughly 30 columns to a smaller set. The SVM model did well, much better than the linear model and random forest. The residual standard error was 1.092263, cutting both previous error values in half. As seen in the figure, the model's guesses for predicted grade are significantly closer to the actual values.

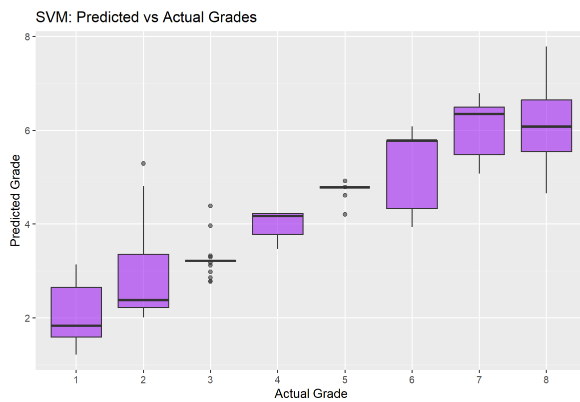

Figure 6: SVM with PCA

3 Conclusions

There was only one usable model, the SVM with PCA model. The clustering lead to no discovery of hidden groups, with much overlap occurring. Both the linear model and the random forest were performed terribly with the given inputs. This proved a larger set of independent variables was needed. The SVM model could potentially be used for real studies. As seen in the figure, for the average student it was fairly good at guessing those values, only struggling with the bottom students and top students.

For future directions, it would be interesting to find the independent variables that have the largest impact on grades. I used all PCAs for the best performing SVM model, but taking apart the best features and just using those might reduce the noise and output better results. In the meantime however, the SVM model is a good model, especially for predicting the students who are going to get grades between 3-7 or DC-CB.

Along with the expansion into identifying key independent variables, an expanded dataset would be ideal. This dataset contained a little over 130 data points, which in the grand scheme is quite small. with a larger data set, clustering may have performed better, or the individual models may have seen more success. This lack of data could be a reason the random forest did better than the linear model, as it could have been overfitting due to the small sample size.